DSCI 310 Final Project: Predicting Energy Consumption Based on Building Attributes

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Background Information

- In recent years, the United States power grid has been struggling to keep up with demand
- The Department of Energy has even come out and said the current usage of the power grid is not sustainable and that upgrades will be needed
- However, these upgrades cannot just be done overnight, it could take years before they are implemented
- In the meantime, we need to find some short term solutions
- If we were able to reliably predict a building’s energy use, this would help out in many ways
    - It would increase grid stability
    - Help out with urban planning
    - Lead to better allocation of resources

Information about the Dataset Used

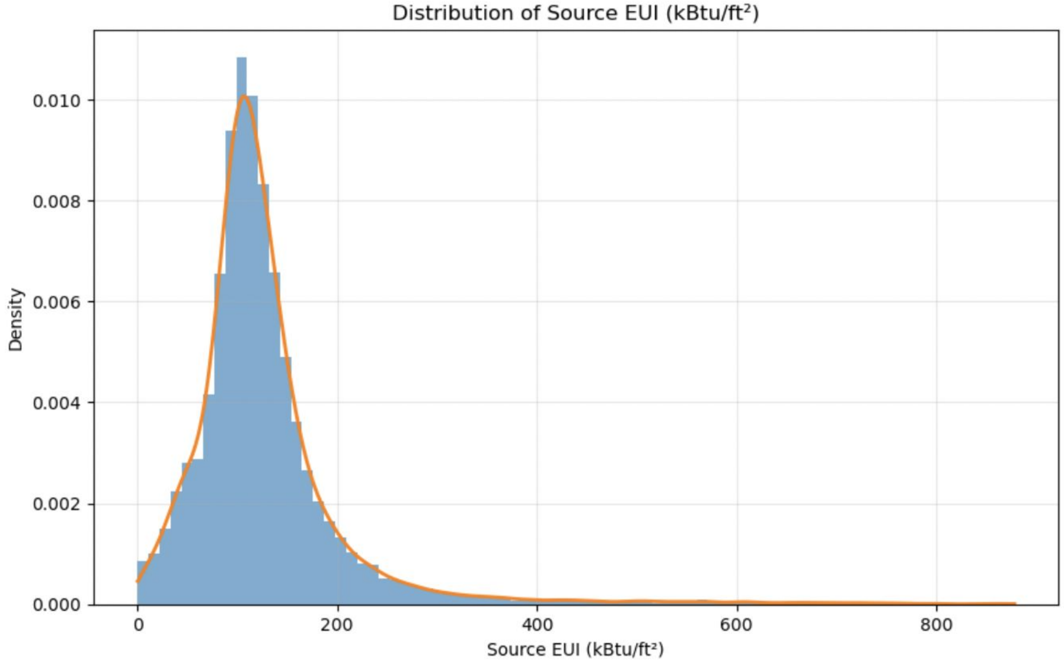
- Contains information for about 30,000 buildings in New York City from 2022
- In total it has 249 columns with a variety of information
  - It has important variables such as building age and number of bedrooms
  - It also has very specific information pertaining to a certain type of building such as a column dedicated to the gross floor area of just bank branches

- Source EUI was the column used as the dependent variable in the regression
  - EUI stands for Energy Use Intensity
  - Source EUI was a measurement of the total energy needed to produce a buildings yearly energy usage in kBtu divided by the total square footage of the building

Model Selection

Gamma regression was chosen for this project due to the Site EUI data following the Gamma distribution

- Skewed to the right
- Variance is not constant
- All values are positive



Information about Gamma Regression

- There are multiple specifications of Gamma regression, for this project log-linked was used
- You are essentially using your predictors to find the natural log of the expected value of Y

$$\log(\mu_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}$$

- Since variance is not constant, a function is used to calculate it

$$\mathrm{Var}(Y_i) = \phi \mu_i^2$$

Cleaning and Preparing the Data

The dataset that was used was fairly messy, and needed to be cleaned

- Outliers were trimmed from the dataset
- Some columns were labeled as containing object and not numerical values, so they needed to be switched
- Problematic predictors were removed from the dataset
- Columns containing a lot of missing values were removed
- Data was split into training, validation, and test sets
- Data was standardized
- VIF filtering was used to detect collinearity
- Columns were quoted since they contained spaces

Results of the Gamma Regression

```
                                                                           coef
err           z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
------------------------------------------------
Intercept                                                               4.7328
002   2491.466      0.000       4.729       4.737
Q("Largest Property Use Type - Gross Floor Area (ft²)")                -0.0489
004    -11.033      0.000      -0.058      -0.040
Q("Year Built")                                                         0.0459
002     23.745      0.000       0.042       0.050
Q("Number of Buildings")                                               -0.0128
002     -6.287      0.000      -0.017      -0.009
Q("Occupancy")                                                          0.0125
002      6.516      0.000       0.009       0.016
Q("Site EUI (kBtu/ft²)")                                                0.4583
002    195.968      0.000       0.454       0.463
Q("Site Energy Use (kBtu)")                                            -0.0225
002     -9.319      0.000      -0.027      -0.018
Q("Net Emissions (Metric Tons CO2e)")                                   0.0900
005     18.665      0.000       0.081       0.099
Q("Number of Active Energy Meters - Not Used to Compute Metrics")       0.0061
002      3.187      0.001       0.002       0.010
```

- The training set was then fitted to the model
- Backward stepwise selection was used to get the predictors that were statistically significant
- After this, the validation set was used
  - MSE: 4994.11
  - MAE: 27.62
- The model was then retrained with both sets, and the test set was used
  - MSE: 4839.40
  - MAE: 27.23
- The prediction intervals were then calculated
  - Average Prediction Interval Width: 137.95
  - Standard Deviation of Prediction Intervals: 128.36
- These results show that the model is sufficient, but not overly great at predicting source EUI

Results of the OLS Regression

	coef	std err	t	P>|t|	[0.025	0.975]
const	127.6441	0.262	486.456	0.000	127.130	128.158
Largest Property Use Type - Gross Floor Area (ft²)	-8.6941	0.613	-14.190	0.000	-9.895	-7.493
Year Built	8.2168	0.267	30.757	0.000	7.693	8.740
Number of Buildings	-1.7833	0.280	-6.363	0.000	-2.333	-1.234
Occupancy	-2.7524	0.265	-10.399	0.000	-3.271	-2.234
Site EUI (kBtu/ft²)	69.6129	0.323	215.515	0.000	68.980	70.246
Site Energy Use (kBtu)	0.0532	0.334	0.159	0.873	-0.601	0.707
Net Emissions (Metric Tons CO2e)	13.3351	0.666	20.018	0.000	12.029	14.641
Number of Active Energy Meters - Not Used to Compute Metrics	-0.6812	0.264	-2.583	0.010	-1.198	-0.164

- A linear model was built to compare to the Gamma regression model
- Followed the same process as the Gamma regression
  - Validation MSE: 1352.99
  - Validation MAE: 24.50
  - Test MSE: 1361.49
  - Test MAE: 24.16
  - Average Prediction Interval Width: 143.36
  - Standard Deviation of Prediction Interval: 0.05
- Surprisingly, the linear model had a much better performance compared to the Gamma GLM

Conclusions and Future Work

- Considering the limitations of the dataset, such as the amount of predictors that had too many NaNs to be put in the model, both models performed well.
- While the OLS had much better MSE values, both models performed relatively similar with respect to their MAE and average prediction interval lengths
  - Variance may not have followed the Gamma variance function a lot of the time
  - Gamma regression penalizes large errors much more than OLS
- Looking into the future, multiple steps can be taken to get better predictions
  - Using interaction effects
  - Using LASSO instead of the backward stepwise solutions
  - Outliers have a huge impact on Gamma regression, could look into using a robust regression model
  - Could also look into using a different type of Gamma regression such as tweedie